

Optimal, Data-Driven Wall Models for Efficient Large Eddy Simulations of Metastable von Kármán Flows

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The von Kármán turbulent swirling flow exhibits intriguing large-scale metastable dynamics, including low-frequency state switching. The study of state switching demands long-duration high-fidelity simulations at high Reynolds numbers that capture the flow generated by the impellers. Blade-resolved Large Eddy Simulations (LES) are computationally prohibitive, limiting access to these slow dynamics. Here, we develop a model for the action of the impellers on the flow using experimental data from Particle Image Velocimetry (PIV) and torque measurements of the von Kármán flow. The impeller-region velocity is parametrized via B-splines and coupled to the LES through momentum forcing. An initial set of B-spline coefficients is inferred using the Optimizing a Discrete Loss (ODIL) framework constrained by the Reynolds-Averaged Navier–Stokes (RANS) equations, PIV measurements in the optically accessible portion of the device, and impeller torque measurements. The coefficients are then refined by the Covariance Matrix Adaptation Evolution Strategy (CMA-ES), which minimizes the discrepancy between the LES time-averaged velocity and torque and their experimental counterparts. Using the data-driven impeller model, we perform long-duration LES of the von Kármán flow. We find that the simulation reproduces the mean flow in the bulk and displays metastable state-switching dynamics. We further show that these metastable states are not axisymmetric and consist of an alternating four-cell flow pattern that slowly rotates around the axis of the cylindrical vessel. The proposed approach provides a practical and computationally efficient route to investigating large-scale dynamics in impeller-driven turbulent flows.

I. INTRODUCTION

Metastable turbulence is a characteristic feature of many high-dimensional chaotic systems across a wide range of natural and industrial settings, including planetary dynamos [1–3], atmospheric flows [4–6], and industrial mixing processes. In these systems, the flow can occupy quasi-persistent states for durations that far exceed the characteristic timescales of the underlying turbulent fluctuations. Understanding the mechanisms behind these transitions is critical for predictability, as they often correspond to high-impact geophysical events or shifts in the efficiency of engineering reactors. A prominent example is found in mid-latitude atmospheric circulation, where the flow intermittently switches between distinct weather regimes such as zonal and blocked states [4, 7, 8]. These metastable regimes represent a significant challenge in non-equilibrium statistical mechanics.

The von Kármán turbulent swirling flow, generated between coaxial counter-rotating impellers in a cylindrical vessel, serves as a prototypical laboratory model for these phenomena. Much like atmospheric regimes, the turbulent von Kármán flow can exhibit multistability and spontaneous symmetry breaking, transitioning from a statistically symmetric two-cell state, through asymmetric two-cell configurations with one enlarged recirculation, to fully asymmetric one-cell attractors [9, 10]. Numerical simulation presents a unique opportunity to probe the physics of these flows beyond what is currently possible with experimental diagnostics alone. Simulation allows for the study of flow structures in regions that are hardly accessible experimentally, such as the immediate vicinity of the impeller blades, and provides the high-fidelity spatio-temporal data needed to characterize the complex three-dimensional flow. Capturing the low-frequency, large-scale dynamics of these bifurcations is computationally demanding, as it requires the simulation of thousands of impeller revolutions to achieve statistical convergence or to observe infrequent state-switching events. Prior numerical simulations of bladed von Kármán flows [11–15] have been restricted to tens of impeller revolutions due to the high computational cost of resolving blade geometry, leaving these slow dynamics inaccessible to numerical investigation. The computational cost of Direct Numerical Simulation (DNS) at the relevant Reynolds numbers is presently prohibitive; consequently, Large Eddy Simulation (LES) is the only practical option for capturing these dynamics. However, the coarse spatial resolution required to enable long-duration simulations prevents an explicit representation of the von Kármán impeller blade geometry. As a result, the action of the impellers must be

modeled. A key challenge for this modeling task is our limited knowledge of the fluid dynamics in the impeller region. Experimental diagnostics such as Particle Image Velocimetry (PIV) provide velocity measurements only in the bulk of the flow. The impeller region, which is the source of momentum injection into the flow, is not directly accessible to PIV measurement.

To address this challenge, a two-stage data-driven optimization strategy is introduced in this paper. The action of the impellers is represented through a parametrized velocity field defined within a prescribed impeller region, rather than through explicit geometric resolution. The impeller-region velocity is expressed as a low-order B-spline parametrization and coupled to the LES through a volumetric forcing of the momentum equations. The B-spline coefficients are determined in two successive steps. In the first step, the Optimizing a DIcrete Loss (ODIL) framework [16, 17] is used to infer an initial set of B-spline coefficients by solving an inverse problem constrained by the Reynolds Averaged Navier–Stokes (RANS) equations, experimental velocity measurements in the bulk, and impeller torque measurements. In the second step, the B-spline coefficients are refined using Covariance Matrix Adaptation Evolution Strategy (CMA-ES) coupled directly to the LES solver, minimizing the discrepancy between the LES time-averaged velocity field and experimental PIV data. This LES-coupled optimization corrects errors introduced by the RANS approximation used in the first step and yields an improved mean-flow prediction. The ODIL-derived coefficients serve two purposes in this second step: i) they initialize the CMA-ES search in a physically consistent region of parameter space, and ii) an LES run with this forcing produces an established flow field that warm-starts each CMA-ES evaluation, reducing the number of impeller revolutions needed to converge time-averaged statistics. We find that this modeling strategy enables long-duration LES of von Kármán flows at high Reynolds number on coarse grids, providing access to slow, large-scale dynamics and metastable state switching that are beyond the reach of blade-resolved simulations at comparable Reynolds numbers and integration times.

Furthermore, the present two-stage optimization strategy can be extended to other problems in fluid mechanics where the forcing region is unresolved or experimentally inaccessible and measurements are sparse. In such settings, ODIL inference constrained by the RANS equations can provide an initialization for a more expensive optimization coupled to LES, whose objective function is difficult to differentiate. Examples include wind turbines and combustion chambers with swirl injectors.

II. METHODS

A. von Kármán flow configuration

The von Kármán flow is established within a closed cylindrical vessel of radius R and height H_V , in which the fluid motion is driven by rotating impellers (Fig. 1). Two identical impellers of height H_I are mounted coaxially at the bottom and at the top of the vessel. The impeller region extends radially from the inner radius R_1 to the outer radius R_2 . The impellers act as the sole source of momentum injection into the flow, producing a turbulent swirling motion confined by the cylindrical walls. In the present work, both impellers rotate at the same angular speed and in opposite directions, $\Omega_1 = -\Omega_2 = \Omega$. The Reynolds number is defined as

$$\text{Re} = \frac{|\Omega|R^2}{\nu}, \quad (1)$$

where ν denotes the kinematic viscosity of the fluid. The geometric parameters of the von Kármán flow device used in this work are given in Table I.

The method is demonstrated and applied to an operating point at $\text{Re} = 300\,000$, with both impellers spinning with the concave side of the blades leading. This operating condition lies in the fully turbulent regime and exhibits large-scale unsteady dynamics, providing a suitable test case for assessing the ability of the method to reproduce both mean-flow statistics and slow dynamical behavior. The PIV velocity fields used for validation were obtained from the experimental campaign of Cappanera et al. [15]. Further details of the experimental setup are given in Appendix A.

TABLE I. Geometric parameters of the von Kármán flow device used to acquire experimental data [15].

R	R_1	R_2	H_V	H_I
100 mm	10 mm	92.5 mm	180 mm	20 mm

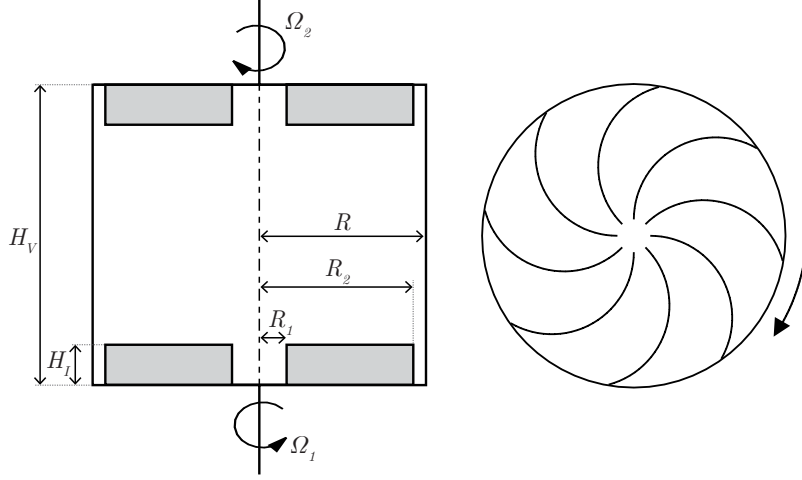


FIG. 1. Von Kármán flow configuration. Left: cylindrical vessel of radius R and height H_V driven by two counter-rotating impellers (angular velocity Ω_1, Ω_2) of height H_I . Right: top view of the curved-blade impeller with inner and outer radii R_1 and R_2 .

B. Large eddy simulation

Large Eddy Simulation (LES) is used to model the turbulent flow in the von Kármán configuration. Only the largest scales are resolved on the computational grid, while the effect of unresolved (subgrid/subfilter) scales on the resolved flow is modeled. Given that compressibility effects are negligible at the operating point, the flow is modeled using the incompressible Navier–Stokes equations:

$$\nabla \cdot \tilde{\mathbf{u}} = 0, \quad (2)$$

$$\frac{\partial \tilde{\mathbf{u}}}{\partial t} + \nabla \cdot (\tilde{\mathbf{u}}\tilde{\mathbf{u}}) = -\frac{1}{\rho}\nabla \tilde{p} - \nabla \cdot \boldsymbol{\tau}^d + \nu \nabla^2 \tilde{\mathbf{u}} + \mathbf{f}, \quad (3)$$

where $\tilde{\cdot}$ denotes a filtered quantity, $\mathbf{u}$ is the velocity vector, ρ is the density, $\tilde{p} = \tilde{p}_* + (1/3)\rho\tau_{kk}$ is the modified pressure where $\tau_{ij} = \widetilde{u_i u_j} - \tilde{u}_i \tilde{u}_j$ is the subgrid-scale tensor, $\tau_{ij}^d = \tau_{ij} - (1/3)\tau_{kk}\delta_{ij}$ is the deviatoric part of the subgrid-scale tensor, ν is the kinematic viscosity, and $\mathbf{f}$ is a body force. The forcing term is used to prescribe the velocity in the impeller region via

$$\mathbf{f} = \alpha\rho(\bar{\mathbf{u}}^{\text{imp}} - \tilde{\mathbf{u}}), \quad (4)$$

where $\bar{\mathbf{u}}^{\text{imp}}$ denotes the target impeller-region velocity. The forcing coefficient is set to $\alpha = \Omega$, equating the relaxation timescale $1/\alpha$ to the characteristic period of impeller rotation. This ensures that momentum is injected on the physically relevant timescale without introducing numerical stiffness. The deviatoric part of the subgrid-scale tensor is approximated using the Smagorinsky-Lilly model [18, 19]

$$\tau_{ij}^d = -2(C_s\Delta)^2|\tilde{\mathbf{S}}|\tilde{S}_{ij}, \quad (5)$$

where C_s is the Smagorinsky coefficient determined dynamically [20] and $|\tilde{\mathbf{S}}| = (2\tilde{S}_{ij}\tilde{S}_{ij})^{1/2}$ is the magnitude of the filtered strain-rate tensor $\tilde{S}_{ij}$. Walls are modeled using a wall model detailed in Appendix C, in which the friction velocity u_τ is determined by solving a one-dimensional boundary-layer equation under the local equilibrium assumption [21–23].

The governing equations are discretized on a structured staggered grid in cylindrical coordinates using second-order accurate centered finite differences and solved using the massively-parallel code NGA [24]. The computational grid is arranged in a cylindrical coordinate system and consists of $N_r \times N_\theta \times N_z = 35 \times 112 \times 62$ cells, where r denotes the radial distance from the axis, θ is the azimuth, and z is the axial coordinate along the axis. Time integration relies on a semi-implicit iterative fractional-step method [25], explicit in the radial and axial directions and implicit in the azimuthal one in order to relax the restrictive stability constraint near the axis. Two sub-iterations per time step are used, yielding nominal second-order accuracy in time for the velocity field. At each sub-iteration, incompressibility is enforced by solving a Poisson equation for the pseudo-pressure and projecting the velocity field onto a divergence-free space.

C. B-spline parametrization of impeller-region velocities

Within the impeller subdomain $\mathcal{D}^{\mathcal{I}}$, the radial and azimuthal velocity components are represented using a low-dimensional B-spline representation in (r, z) . This reduces the dimensionality of the parameter space to a small set of coefficients, which parametrize the impeller-region velocity in both the ODIL inference and CMA-ES optimization steps. Only the radial and azimuthal components of the impeller-region velocity are parametrized; the axial component is determined implicitly through the continuity equation. Let $\{B_a^{(z)}(z)\}_{a=1}^{n_z}$ and $\{B_b^{(r)}(r)\}_{b=1}^{n_r}$ denote clamped, uniformly spaced B-spline basis functions of degree p defined on the axial interval $|z| \in [H_V/2 - H_I, H_V/2]$ and the radial interval $r \in [R_1, R_2]$ spanned by the blades. For $\eta \in \{r, \theta\}$ we define

$$\bar{u}_\eta^{\text{imp}}(r, z) = \sum_{a=1}^{n_z} \xi_{\eta,a} B_a^{(z)}(z) + \sum_{b=1}^{n_r} \chi_{\eta,b} B_b^{(r)}(r), \quad (6)$$

where $\xi_\eta \in \mathbb{R}^{n_z}$ and $\chi_\eta \in \mathbb{R}^{n_r}$ are vectors of axial and radial spline coefficients, treated as optimization variables. We use quadratic splines ($p = 2$) with $n_z = 3$ axial basis functions and $n_r = 3$ radial basis functions. Because of the symmetry, the same axial splines at $z_{\text{mir}} = -z$ are used with the same coefficients for both regions. The four coefficient vectors are stacked into a single vector

$$\mathbf{c} = (\xi_r, \chi_r, \xi_\theta, \chi_\theta) \in \mathbb{R}^{2(n_z+n_r)}, \quad (7)$$

forming the unknown parameters to be determined with ODIL and CMA-ES.

D. ODIL-based inference

The ODIL-based inference retrieves an effective representation of the mean impeller-region velocities that is consistent with the RANS equations and the available experimental data. Impeller-region velocities $\bar{\mathbf{u}}^{\text{imp}}$, parametrized through the coefficient vector $\mathbf{c}$ (Eq. (7)), are inferred by solving an inverse problem. A key challenge is that experimental data from PIV provide velocity measurements only in the bulk flow, away from the impeller blades. Thus, $\bar{\mathbf{u}}^{\text{imp}}$ is retrieved using solutions to the RANS equations with the objective that the mean velocity field from RANS matches available PIV data. The Optimizing a DIscrete Loss (ODIL) framework [16, 17] is used to jointly minimize the discrete RANS residual and a data-misfit term quantifying the discrepancy between the RANS solution and the experimental measurements. The resulting set of coefficients $\mathbf{c}^{(0)}$ serves two purposes in the subsequent CMA-ES optimization. First, it initializes the CMA-ES search in a physically consistent region of the parameter space. Second, an LES run with this forcing produces an established flow field that is already close to the target state; this flow field is used to warm-start the LES evaluations within CMA-ES, so that fewer impeller revolutions are required for time-averaged statistics to converge at each candidate evaluation, substantially reducing the computational cost of the optimization.

a. RANS equations The RANS equations for a statistically stationary and incompressible flow read as follows

$$\nabla \cdot \bar{\mathbf{u}} = 0, \quad (8)$$

$$\nabla \cdot (\bar{\mathbf{u}}\bar{\mathbf{u}}) + \frac{1}{\rho} \nabla \tilde{p} + \nabla \cdot \boldsymbol{\tau}_R^d - \nu \nabla^2 \bar{\mathbf{u}} = 0, \quad (9)$$

where $\bar{\cdot}$ denotes a temporal mean quantity, $\mathbf{u}$ is the velocity, ρ is the density, $\tilde{p} = \bar{p} + 2/3 \rho k$ is the modified pressure, ν is the kinematic viscosity and $\boldsymbol{\tau}_R^d$ is the deviatoric Reynolds stress tensor which is modeled through an eddy-viscosity closure $\boldsymbol{\tau}_R^d = -\nu_t (\nabla \bar{\mathbf{u}} + \nabla \bar{\mathbf{u}}^T)$. The turbulent viscosity ν_t is modeled using the $k - \varepsilon$ model

$$\nu_t = C_\nu k^2 / \varepsilon, \quad (10)$$

$$\nabla \cdot (k \bar{\mathbf{u}}) - \nabla \cdot ((\nu + \nu_t / \sigma_k) \nabla k) - P_k / \rho + \varepsilon = 0, \quad (11)$$

$$\nabla \cdot (\varepsilon \bar{\mathbf{u}}) - \nabla \cdot ((\nu + \nu_t / \sigma_\varepsilon) \nabla \varepsilon) - C_1 \varepsilon P_k / \rho k + C_2 \varepsilon^2 / k = 0, \quad (12)$$

where k is the turbulent kinetic energy, P_k its production and ε its dissipation. The model constants are $C_\nu = 0.09$, $\sigma_k = 1.0$, $\sigma_\varepsilon = 1.3$, $C_1 = 1.44$ and $C_2 = 1.92$ [26]. Walls are modeled using a friction factor defined as a function of the local Reynolds number [27].

The RANS equations are discretized in axisymmetric cylindrical coordinates (r, z) using second-order central finite differences on a staggered Marker-and-Cell (MAC) grid. The computational grid comprises $N_r \times N_z = 70 \times 62$ cells in the radial and axial directions, respectively. The radial and axial velocity components $(\bar{u}_r, \bar{u}_z)$ are stored on cell faces, while the azimuthal velocity $\bar{u}_\theta$, modified pressure $\tilde{p}$, and turbulence variables (k, ε) are stored at cell centers. The discretization leads to a nonlinear system of algebraic equations for the discrete unknowns.

b. Free parameters We decompose the computational domain as $\mathcal{D} = \mathcal{D}^{\mathcal{I}} \cup \mathcal{D}^{\mathcal{F}}$ with $\mathcal{D}^{\mathcal{F}} := \mathcal{D} \setminus \mathcal{D}^{\mathcal{I}}$ where $\mathcal{D}^{\mathcal{I}}$ denotes the impeller subdomain and $\mathcal{D}^{\mathcal{F}}$ its complement. The optimization vector comprises both grid-based RANS unknowns and the spline coefficients defining the impeller kinematics. The axial velocity component $\bar{u}_z$ is treated as a grid unknown over the entire domain $\mathcal{D}$, including the impeller region. The radial and azimuthal components are treated differently. In the free-velocity region $\mathcal{D}^{\mathcal{F}}$, $\bar{u}_r$ and $\bar{u}_\theta$ are standard grid degrees of freedom. In contrast, within the impeller subdomain $\mathcal{D}^{\mathcal{I}}$ these components are not optimized pointwise. Instead, they are prescribed through the spline parametrization $\bar{u}_r^{\text{imp}}(\cdot; \mathbf{c})$ and $\bar{u}_\theta^{\text{imp}}(\cdot; \mathbf{c})$, whose coefficient vector $\mathbf{c}$ constitutes a low-dimensional control variable. The modified pressure $\tilde{p}$ and turbulence-model variables (k, ε) are optimized throughout $\mathcal{D}$.

Accordingly, the optimization variables can be summarized as:

$$\mathbf{v} = \left(\bar{u}_r|_{\mathcal{D}^{\mathcal{F}}}, \bar{u}_z|_{\mathcal{D}}, \bar{u}_\theta|_{\mathcal{D}^{\mathcal{F}}}, \tilde{p}|_{\mathcal{D}}, k|_{\mathcal{D}}, \varepsilon|_{\mathcal{D}}, \mathbf{c} \right), \quad (13)$$

where the restrictions indicate the collection of interior MAC-grid degrees of freedom in the corresponding subdomain.

c. Loss functional The ODIL objective is defined as a weighted least-squares norm of the discrete residuals of the governing equations together with the data-misfit terms. The combination of the two terms can also be formulated in a Bayesian framework as a prior (the equation residuals) and a likelihood term (the data term) [28]. Let $\mathbf{r}_m$, $\mathbf{r}_c$, $\mathbf{r}_k$ and $\mathbf{r}_\varepsilon$ denote the discrete momentum, continuity, turbulent kinetic energy and dissipation residual vectors, respectively. The continuity equation (Eq. (8)) and the k - ε transport equations (Eqs (11) and (12)) are enforced throughout the entire domain $\mathcal{D}$, whereas the momentum equations for $(\bar{u}_r, \bar{u}_\theta, \bar{u}_z)$ (Eq. (9)) are enforced only in the free-velocity region $\mathcal{D}^{\mathcal{F}}$. Inside the impeller region $\mathcal{D}^{\mathcal{I}}$, the momentum equations are not enforced: $\bar{u}_r$ and $\bar{u}_\theta$ are prescribed by the spline parametrization, and $\bar{u}_z$ adjusts to satisfy continuity. The resulting loss functional reads

$$\mathcal{J}(\mathbf{v}) = \frac{1}{N_m} \|\mathbf{r}_m|_{\mathcal{D}^{\mathcal{F}}}\|_2^2 + \frac{1}{N_c} \|\mathbf{r}_c|_{\mathcal{D}}\|_2^2 + \frac{1}{N_k} \|\mathbf{r}_k|_{\mathcal{D}}\|_2^2 + \frac{1}{N_\varepsilon} \|\mathbf{r}_\varepsilon|_{\mathcal{D}}\|_2^2 + w_{\text{PIV}} \frac{1}{N_{\text{PIV}}} \|\mathbf{r}_{\text{PIV}}\|_2^2 + w_Q |r_Q|^2, \quad (14)$$

where $\mathbf{r}_{\text{PIV}}$ and r_Q denote the mismatches between the computed solution and the available measurements for the velocity field (PIV) and the impeller torque, respectively. The weights w_{PIV} and w_Q control the relative contributions of the two data-misfit terms to the loss functional. The data-misfit weights are set to $w_{\text{PIV}} = 2$ and $w_Q = 1 \times 10^{-3}$, balancing fidelity to the experimental measurements with enforcement of the governing Partial Differential Equation (PDE) residuals from RANS.

E. CMA-ES optimization

The ODIL-inferred set of coefficients $\mathbf{c}^{(0)}$ provides a physically consistent starting point, but it is subject to the modeling error inherent in the RANS approximation. In order to correct this residual error, the B-spline coefficients are refined using the Covariance Matrix Adaptation Evolution Strategy (CMA-ES) [29], a derivative-free evolutionary optimization strategy that iteratively adapts a multivariate Gaussian distribution over the parameter space to minimize a black-box objective.

The objective function combines the discrepancy between the LES time-averaged velocity field and the experimental PIV data, the discrepancy between the LES time-averaged impeller torque and the experimentally measured torque, and a physics-based penalty on the target impeller-region velocity field,

$$\mathcal{F}(\mathbf{c}) = \frac{1}{N_p} \|\langle \tilde{\mathbf{u}} \rangle_T(\mathbf{c}) - \bar{\mathbf{u}}^e\|_2^2 + w_{Q'} (\langle Q \rangle_T(\mathbf{c}) - \bar{Q}^e)^2 + w_\phi \phi(\mathbf{c}) \quad (15)$$

where $\langle \cdot \rangle_T$ denotes a time average over T impeller revolutions, $\mathbf{c}$ is the B-spline coefficient vector defined in Eq. (7), $\langle Q \rangle_T(\mathbf{c})$ is the mean torque transmitted to the fluid by the impeller model and $\bar{Q}^e$ is the corresponding experimentally measured torque. The penalty weight $w_{Q'}$ is set to $w_{Q'} = 1 \times 10^{-2}$. Another penalty term $w_\phi \phi(\mathbf{c})$ encodes two physical constraints on the B-spline target velocity field $\tilde{\mathbf{u}}^{\text{imp}}(\mathbf{c})$

$$\phi(\mathbf{c}) = \phi_r(\mathbf{c}) + \phi_\theta(\mathbf{c}) \text{ with} \quad (16)$$

$$\phi_r(\mathbf{c}) = \frac{1}{|\mathcal{D}^{\mathcal{I}}|} \int_{\mathcal{D}^{\mathcal{I}}} \max(0, -\bar{u}_r^{\text{imp}}(\mathbf{c}))^2 d\mathcal{D} \text{ and} \quad (17)$$

$$\phi_\theta(\mathbf{c}) = \frac{1}{|\mathcal{D}^{\mathcal{I}}|} \int_{\mathcal{D}^{\mathcal{I}}} \max(0, -\bar{u}_\theta^{\text{imp}}(\mathbf{c}) \text{sgn}(\Omega_1))^2 d\mathcal{D}, \quad (18)$$

where Ω_i is the angular velocity of the local impeller. The term ϕ_r penalizes inward radial velocity, enforcing centrifugal pumping; ϕ_θ penalizes azimuthal velocity opposing the impeller direction of rotation. The penalty weight is set to $w_\phi = 0.1$. Without this penalty, fitting the sparse PIV data alone can drive the B-spline coefficients toward target velocity fields that locally violate the expected impeller flow physics (e.g., inward radial velocity or rotation-opposing azimuthal velocity), which in turn can destabilize or corrupt the LES near the impeller region. No constraint is imposed on the axial component, as it is not an independent B-spline degree of freedom but is instead recovered from the continuity equation.

At each CMA-ES iteration, a population of candidate coefficient vectors is sampled from the current Gaussian distribution, each candidate is evaluated by running the LES and computing $\mathcal{F}$, and the distribution mean and covariance matrix are updated to favor regions of lower objective value. The population size λ follows the standard CMA-ES default, $\lambda = 4 + \lfloor 3 \ln n_{\text{dim}} \rfloor$, which yields $\lambda = 11$ individuals per generation for the $n_{\text{dim}} = 12$ B-spline parametrization. The distribution mean is initialized with $\mathbf{c}^{(0)}$ from ODIL, so that the search begins from a physically consistent region of the parameter space. The covariance matrix is initialized to the identity, with initial step size $\sigma_{\text{CMAES}}^{\text{init}} = 0.1$, corresponding to an isotropic initial search distribution. Each candidate LES evaluation is warm-started from the established flow field obtained by running the LES with the ODIL forcing prior to the CMA-ES iterations; since the flow does not need to spin up from rest, fewer impeller revolutions are required to converge the time-averaged velocity field, reducing the cost per function evaluation. Specifically, each evaluation discards the first 5 impeller revolutions as a burn-in period and then averages the velocity field over the subsequent $T = 35$ revolutions to compute $\mathcal{F}$. This averaging window was chosen so that the statistical noise in $\mathcal{F}$ arising from finite-time averaging remains substantially smaller than the differences in $\mathcal{F}$ that CMA-ES must resolve between competing candidates. The optimized coefficient vector $\mathbf{c}^*$ is then used to define the impeller-region forcing for long-duration LES of the von Kármán flow.

III. RESULTS AND DISCUSSION

A. ODIL-based inference of impeller-region velocity fields

Prior to solving the inverse problem, a forward RANS computation is performed in which a prescribed azimuthal velocity $\bar{u}_\theta|_{\mathcal{D}^\pm}(r) = \Omega r$ is imposed within the impeller regions $\mathcal{D}^\pm$. This provides an initial velocity field that serves as the starting point for the subsequent ODIL-based optimization. The inverse procedure then relaxes this constraint and infers the effective velocity distribution that best satisfies both the governing equations and the experimental data.

The nonlinear least-squares problem is solved using the Gauss-Newton algorithm described in Appendix B. Convergence is achieved after approximately 1 600 iterations, as evidenced by the monotonic decay and subsequent stabilization of the objective functional and of the individual residual norms (Fig. 2). No residual stagnation or oscillatory behavior is observed. The ODIL-based optimization then yields an effective velocity field within the prescribed impeller regions that represents the net action of the rotating blades on the flow (Fig. 3). The spatial distribution of the error remains weak throughout the bulk, with slightly larger deviations localized near the outer radial boundary and in the vicinity of the impeller edges, where strong velocity gradients are present. The inferred velocity components exhibit spatial distributions that are consistent with the expected impeller-induced motion. The azimuthal component reflects the primary role of the blades in injecting angular momentum into the flow, increasing with radius and reaching a maximum near the outer impeller radius, where blade-tip effects are strongest. The radial component contributes to the cross-stream transport needed to sustain axial pumping and the global recirculation. The axial velocity component, retrieved from the continuity constraint, represents the axial pumping implied by the inferred cross-stream motion. The ODIL solution provides a physically consistent starting point for the CMA-ES optimization, which corrects residual errors introduced by the RANS approximation.

B. LES with ODIL-inferred impeller-region velocity parametrization

The ODIL-inferred coefficient vector $\mathbf{c}^{(0)}$ (Eq. (7)) is used to define the impeller forcing in LES of the von Kármán flow. The mean velocity field obtained from this LES is compared against the experimental PIV data (Fig. 4). The LES reproduces the large-scale structure of the flow: the two counter-rotating recirculation cells are present, the azimuthal velocity magnitude and its spatial distribution are consistent with the measurements, and the meridional circulation correctly connects the impeller regions to the bulk. The global Root Mean Square Error (RMSE) between the LES and the experimental data is $0.05 \Omega R$, and the RSE map shows that errors are largest near the inner axial edge of the impeller and the outer radial boundary, where strong velocity gradients are present, while the bulk of the

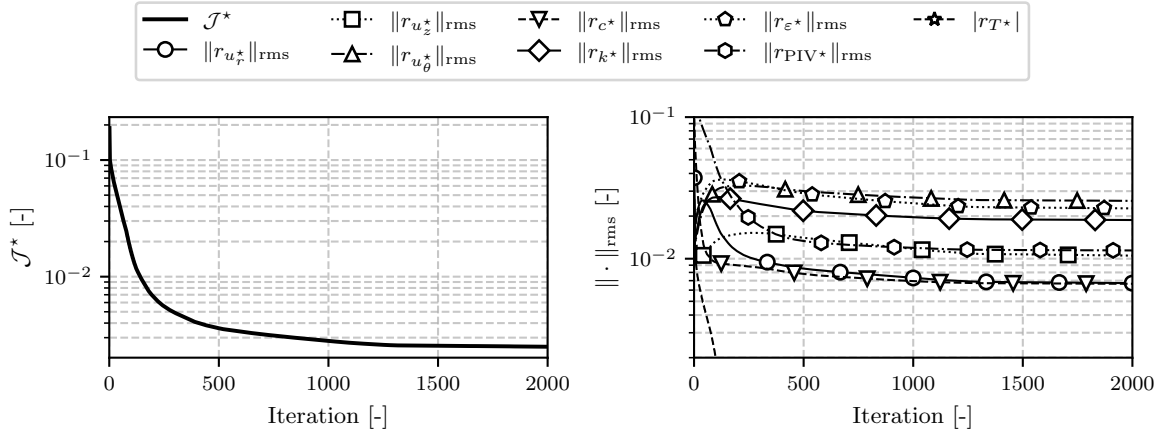


FIG. 2. Convergence history of the RANS-ODIL flow reconstruction. Left: evolution of the objective function constructed from nondimensional residuals $\mathcal{J}^*$. Right: evolution of the RMS of the residual components (Eq. (14)) with $\|\cdot\|_{\text{RMS}} = (N^{-1} \sum_i^N (\cdot)_i^2)^{1/2}$. The monotonic decay of the objective and stabilization of the residual norms indicate convergence of the coupled RANS-data optimization.

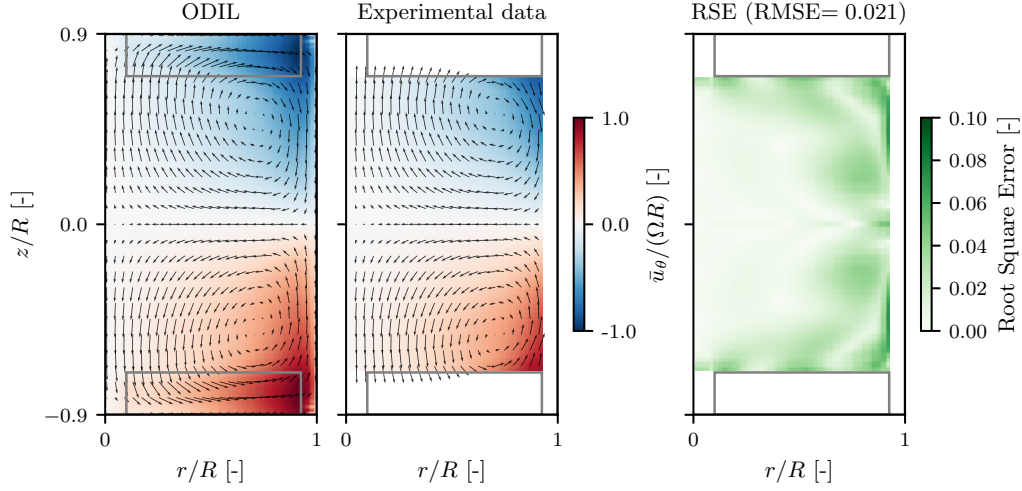


FIG. 3. Comparison of the mean velocity field in the meridional plane. From left to right: ODIL reconstruction, experimental measurements, and Root Square Error (RSE) between ODIL and experiments. Fields are shown in the $(r/R, z/R)$ coordinates, with in-plane velocity vectors superimposed on the ODIL and experimental mean azimuthal velocity $\bar{u}_\theta/(\Omega R)$.

domain is well captured. This result demonstrates that the B-spline parametrization initialized through a RANS-based inverse problem solved with ODIL already provides a physically consistent and quantitatively accurate representation of the impeller forcing. The CMA-ES optimization is subsequently performed to correct the modeling error introduced by the RANS approximation used in the ODIL-based inference.

C. CMA-ES optimization

Starting from the ODIL-inferred coefficient vector $\mathbf{c}^{(0)}$ and the established flow field from the corresponding LES, the CMA-ES optimization is run to refine the B-spline parametrization. The population mean of the objective function $\mathcal{F}(\mathbf{c})$ (Eq. (15)) decreases rapidly over the first generations and stabilizes after approximately 60 generations (Fig. 5), indicating convergence of the optimization.

The mean velocity field obtained from the LES with the CMA-ES optimized coefficients $\mathbf{c}^*$ is compared against the experimental PIV data (Fig. 6). The RMSE is reduced from $0.050 \Omega R$ (ODIL) to $0.027 \Omega R$ (CMA-ES), a 46% improvement. The two counter-rotating recirculation cells are well reproduced, the azimuthal velocity and its radial variation are in close agreement with the measurements, and the meridional velocity vectors show the correct large-

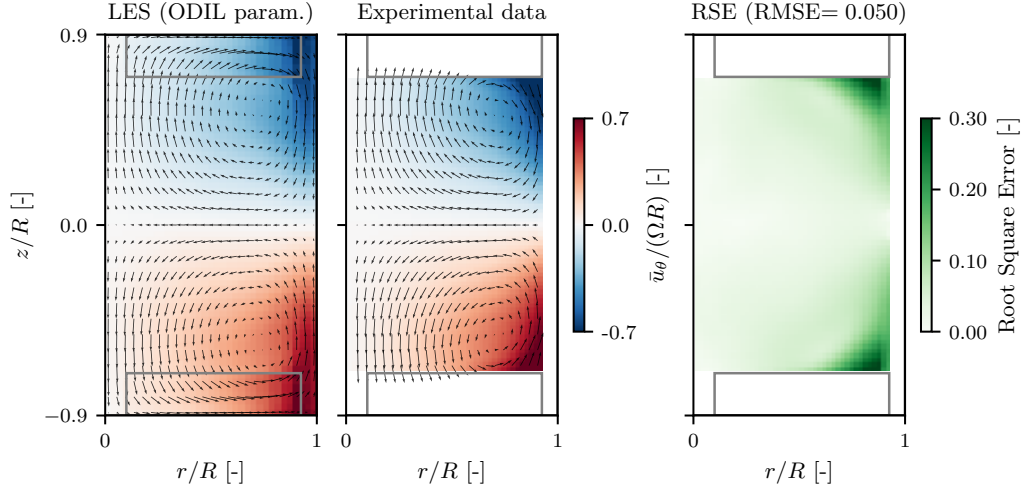


FIG. 4. Comparison of the mean velocity field in the meridional plane. From left to right: LES with ODIL-inferred impeller-region velocity parametrization $\mathbf{c}^{(0)}$, experimental measurements, and Root Square Error (RSE) between LES and experiments. Fields are shown in the $(r/R, z/R)$ coordinates, with in-plane velocity vectors superimposed on the LES and experimental mean azimuthal velocity $\bar{u}_\theta/(\Omega R)$.

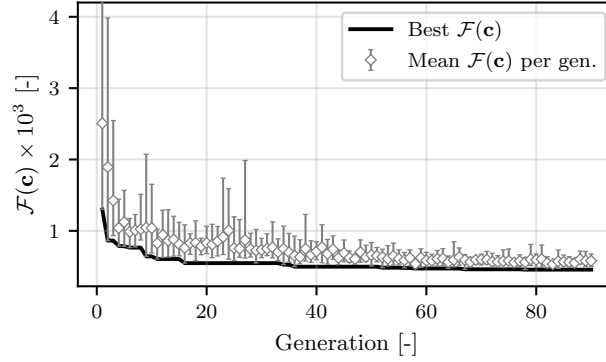


FIG. 5. Evolution of the CMA-ES objective function $\mathcal{F}(\mathbf{c})$ (Eq. (15)) over optimization generations. The solid line shows the best objective value found up to each generation (overall best). Diamond markers show the mean $\mathcal{F}(\mathbf{c})$ within each generation, with error bars spanning the minimum and maximum values.

scale circulation topology throughout the bulk.

The improvement is further confirmed by the radial profiles of all three mean velocity components at three axial positions (Fig. 7). The CMA-ES solution (solid lines) consistently lies closer to the experimental data (symbols) than the ODIL baseline (dash-dotted lines) across all components and axial positions. The azimuthal velocity profiles show the largest improvement, with the optimized solution accurately capturing the radial variation toward the outer wall. The radial and axial velocity profiles show modest improvement. Remaining discrepancies are localized near the outer radial boundary and the impeller edges, where the steep velocity gradients are expected to be most sensitive to the absence of explicit blade geometry. Importantly, the accuracy achieved in the bulk is sufficient to sustain the correct large-scale circulation patterns and to enable the emergence of intrinsic low-frequency dynamics, which are the objective of the present modeling work and are examined in the following section.

D. Metastable flow states and large-scale transitions

We now study the dynamics of the flow obtained from the LES with the CMA-ES optimized coefficient vector $\mathbf{c}^*$. By running the LES for a long duration, we observe metastable large-scale flow states and intermittent transitions between them (Fig. 8). To characterize these metastable states, an order parameter is constructed from the azimuthally and axially averaged axial velocity sampled in the near-wall annular region $0.90 \leq r/R \leq 0.95$, $|z| < z_{\text{lim}}$, where $z_{\text{lim}}/R = 0.5$ excludes the impeller forcing regions. For each azimuthal plane at angular position θ_k ($k = 1, \dots, N_\theta$),

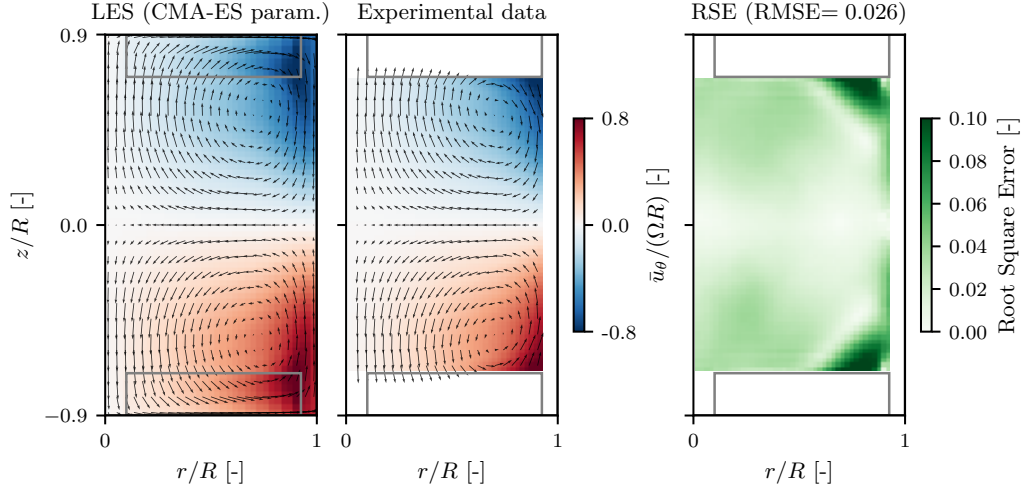


FIG. 6. Comparison of the mean velocity field in the meridional plane. From left to right: LES with CMA-ES optimized impeller-region velocity parametrization $\mathbf{c}^*$, experimental measurements, and Root Square Error (RSE) between LES and experiments. Fields are shown in the $(r/R, z/R)$ coordinates, with in-plane velocity vectors superimposed on the LES and experimental mean azimuthal velocity $\bar{u}_\theta/(\Omega R)$.

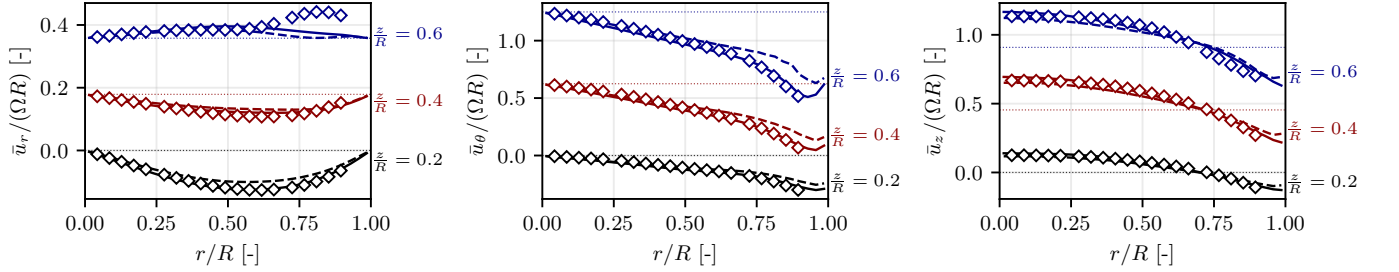


FIG. 7. Radial profiles of the mean velocity components $\bar{u}_r/(\Omega R)$ (left), $\bar{u}_\theta/(\Omega R)$ (center), and $\bar{u}_z/(\Omega R)$ (right) at three axial positions z/R . Profiles are offset by the local zero-velocity baseline for clarity (cascade representation). Symbols: PIV experiments; dash-dotted lines: LES with ODIL-inferred impeller-region velocity parametrization $\mathbf{c}^{(0)}$; solid lines: LES with CMA-ES-optimized impeller-region velocity parametrization $\mathbf{c}^*$.

the local indicator is defined as

$$q_{u_z}^{(k)}(t) = \frac{\int_{|z| < z_{\text{lim}}} \int_{0.90R}^{0.95R} u_z(r, z, \theta_k, t) dA}{\int_{|z| < z_{\text{lim}}} \int_{0.90R}^{0.95R} dA}, \quad (19)$$

where $dA = r dr dz$ is the meridional area element. Near the outer wall, the axial velocity is directly related to the large-scale meridional circulation: a positive (negative) q_{u_z} indicates a net upward (downward) axial flux at the periphery, which discriminates between the two asymmetric recirculation states. This near-wall quantity thus provides a geometrically local and computationally inexpensive proxy for the symmetry-breaking dynamics. The metastable states are highlighted using an Exponential Moving Average (EMA) of the order parameter. The EMA is defined as

$$\hat{\varphi}(t) = \frac{1}{T_s} \int_{-\infty}^t \exp\left(-\frac{t-\tau}{T_s}\right) \varphi(\tau) d\tau, \quad (20)$$

for a variable φ . T_s is the averaging time scale and is chosen as $T_s = 5$ impeller revolutions, based on the decorrelation time of the order parameter. This choice ensures that the averaging filters out fast turbulent fluctuations while preserving the slow, large-scale dynamics associated with metastable flow states. The system is assigned to state A (respectively state B) at time t if the EMA-filtered dimensionless order parameter $\hat{q}_{u_z}^{(k)}/\Omega R$ has remained continuously above a threshold $\gamma_A = 0.1$ (respectively below $\gamma_B = -0.1$) over a trailing dwell window of duration $T_d = 5$ impeller revolutions; otherwise the state is left undefined. This dwell-time requirement prevents brief threshold crossings due to residual turbulent fluctuations from being misclassified as state transitions, so that only genuine, sustained

excursions into either macroscopic configuration are labeled. While fast turbulent fluctuations are visible at short time scales, the signal also exhibits slow variations characterized by extended plateaus separated by relatively rapid transitions. These plateaus correspond to quasi-persistent flow states, indicating that the system remains trapped in distinct large-scale configurations for durations that far exceed the characteristic turbulent time. The Probability Density Function (PDF) of $\hat{q}_{u_z}$ is bimodal (Fig. 9), reflecting the coexistence of two preferred macroscopic states. Such behavior is a hallmark of metastable dynamics in the von Kármán configuration and is qualitatively consistent with experimental observations [10, 30].

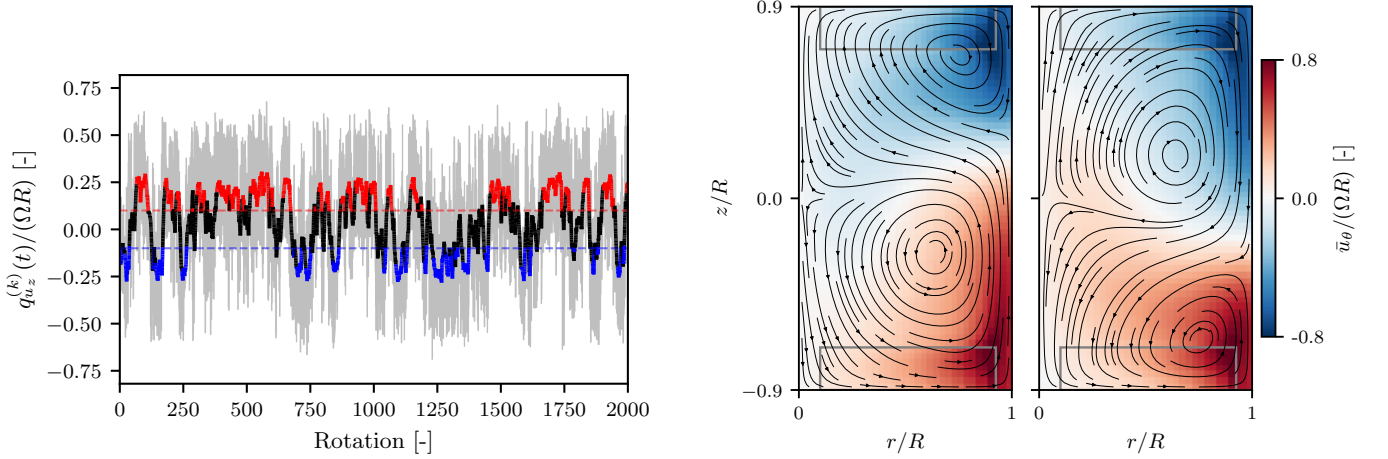


FIG. 8. Time evolution and representative flow states of the von Kármán flow at angular position $\theta_k = \pi/4$. Left: Time series of the dimensionless order parameter $q_{u_z}^{(k)}(t)/\Omega R$ (gray) together with its exponential moving average colored by the state of the system (red: state A, black: undefined, blue: state B). Right: Streamlines of the mean meridional velocity field (u_r, u_z) conditioned on the state of the system (A/B), illustrating two metastable flow states. In both cases, the flow consists of two recirculation cells of unequal strength, with the dominant circulation located either above or below the mid-plane.

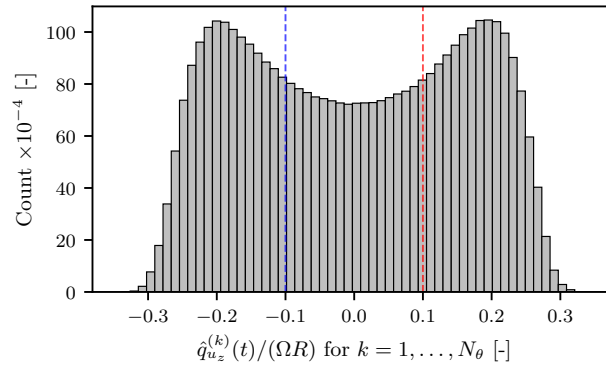


FIG. 9. PDF of the EMA-filtered dimensionless order parameter $\hat{q}_{u_z}^{(k)}(t)/\Omega R$ computed from long-duration LES (4,500 rotations) and assembled by sampling at all $k = 1, \dots, N_{\theta}$. The bimodal distribution reflects the coexistence of two symmetry-related metastable states with opposite dominance of the large-scale circulation.

Both metastable states are asymmetric and consist of two recirculation cells of unequal size and intensity (Fig. 8). In one case, a dominant large-scale circulation cell occupies the upper part of the vessel, while a smaller secondary cell persists near the bottom impeller. In the other case, the roles of the two cells are reversed, with the dominant circulation located below the mid-plane and a weaker cell remaining above. These two asymmetric configurations correspond to symmetry-related metastable states of the von Kármán flow. Their coexistence, together with the intermittent transitions observed in the time evolution of $\hat{q}_{u_z}^{(k)}(t)$, is indicative of spontaneous symmetry breaking of the mean flow topology. An animation of a meridional plane showing the EMA-filtered velocity field is provided as supplementary material and illustrates the transition between the two metastable states.

To quantify the characteristic lifetime of each metastable state, the residence time is computed, i.e., the length of each uninterrupted interval during which the system remains classified in state A or state B, pooled over all azimuthal

planes. The resulting distributions of residence times (Fig. 10) decay approximately exponentially, suggestive of a memoryless escape process, consistent with noise-driven transitions between metastable states rather than a sharp deterministic escape time. An exponential-decay fit to each distribution yields residence times of $\tau_A \approx 15.6$ and $\tau_B \approx 15.7$ impeller revolutions for states A and B, respectively, quantifying the persistence of the large-scale asymmetric circulation before a spontaneous transition occurs. The near-equality of τ_A and τ_B is consistent with the symmetry-related nature of the two metastable states when the two impellers counter-rotate at the same frequency.

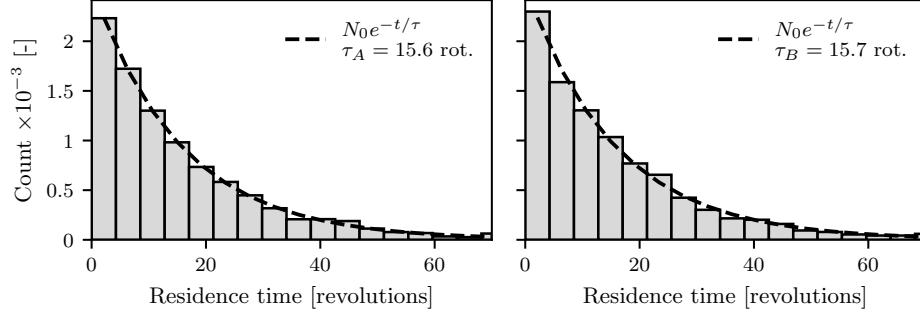


FIG. 10. Histogram of residence times in state A (left) and state B (right), obtained from the EMA-filtered order parameter $\hat{q}_{u_z}^{(k)}(t)$ and pooled over all azimuthal planes $k = 1, \dots, N_\theta$. Dashed lines show exponential-decay fits to each distribution, with fitted residence times reported in the legend.

The ability of the LES to sustain these asymmetric states and transition between them shows that the large-scale metastable dynamics characteristic of the von Kármán configuration are captured by the present modeling approach. Importantly, the transitions between states occur spontaneously, without any imposed time-dependent forcing, demonstrating that the present impeller model allows the intrinsic low-frequency dynamics of the system to emerge naturally.

1. Azimuthal structure of the metastable states

The analysis so far has focused on a single angular position. However, the LES resolves the full azimuthal extent of the flow, which allows the azimuthal organization of flow states to be characterized. To this end, the equal-time Pearson correlation coefficient of the EMA-filtered order parameter $\hat{q}_{u_z}^{(k)}$ is computed between all pairs of azimuthal planes θ_j, θ_k , $j, k = 1, \dots, N_\theta$ (Fig. 11). Diametrically opposite planes ($|\theta_j - \theta_k| = \pi$) are positively correlated, whereas planes separated by a quarter turn ($|\theta_j - \theta_k| = \pi/2$) are anti-correlated. This pattern is the signature of a dominant azimuthal wavenumber $m = 2$: the metastable states identified from the meridional-plane statistics are not axisymmetric but consist of two pairs of counter-rotating recirculation cells distributed around the circumference of

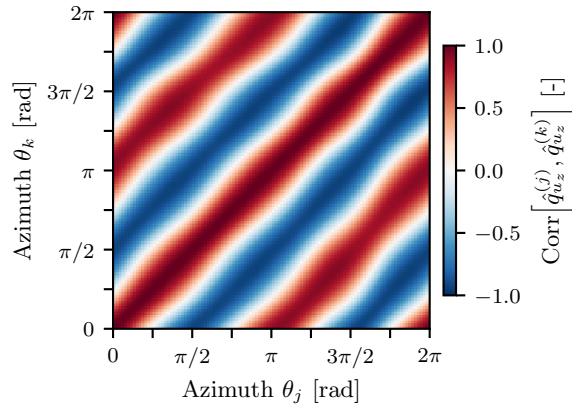


FIG. 11. Equal-time Pearson correlation matrix of the EMA-filtered order parameter $\hat{q}_{u_z}^{(k)}$ between all pairs of azimuthal planes θ_j, θ_k , $j, k = 1, \dots, N_\theta$. Diametrically opposite planes ($|\theta_j - \theta_k| = \pi$) are positively correlated. Conversely, planes separated by a quarter turn ($|\theta_j - \theta_k| = \pi/2$) are anti-correlated, indicating that the metastable states have a dominant azimuthal wavenumber $m = 2$.

the vessel.

This wavenumber-2 organization is confirmed directly in physical space in Fig. 12, which shows, for three distinct instants, an unrolled cylindrical section near the outer wall together with a cross-sectional cut normal to the cylinder axis, at the mid-plane of the vessel (hereafter, the mid-height cut). In each case, the mid-height cut exhibits a clear four-cell pattern, with axial velocity of alternating sign in azimuthal sectors of extent $\pi/2$: positively correlated between diametrically opposite sectors and anti-correlated between sectors separated by a quarter turn, consistent with the correlation analysis of Fig. 11. The three instants selected have their positive order-parameter lobe centered at three different azimuthal positions ($\theta = 0, \pi/4$ and $\pi/2$, respectively), showing that the four-cell structure is not pinned to a fixed orientation: the metastable states are degenerate with respect to a rotation of the pattern about the cylinder axis, and the wavenumber-2 organization persists regardless of its azimuthal phase.

The azimuthal phase of the four-cell pattern is not static: inspection of the corresponding animation (provided as supplementary material) shows that the structure typically drifts slowly around the circumference, rotating in either direction in an apparently random fashion, rather than remaining locked at a fixed θ . Transitions between metastable states, in the sense of Fig. 8, most often proceed through this gradual azimuthal drift. In rarer instances, however, the four-cell organization instead collapses locally before reforming, without a clearly defined rotation; Fig. 13 shows an example of such an instant, for which the state of the system is ambiguous. In this case, the mid-height cut no longer exhibits two distinct pairs of counter-rotating cells but instead a single dominant recirculation, reflecting a momentary loss of the wavenumber-2 organization as the flow reorganizes from one metastable configuration to another.

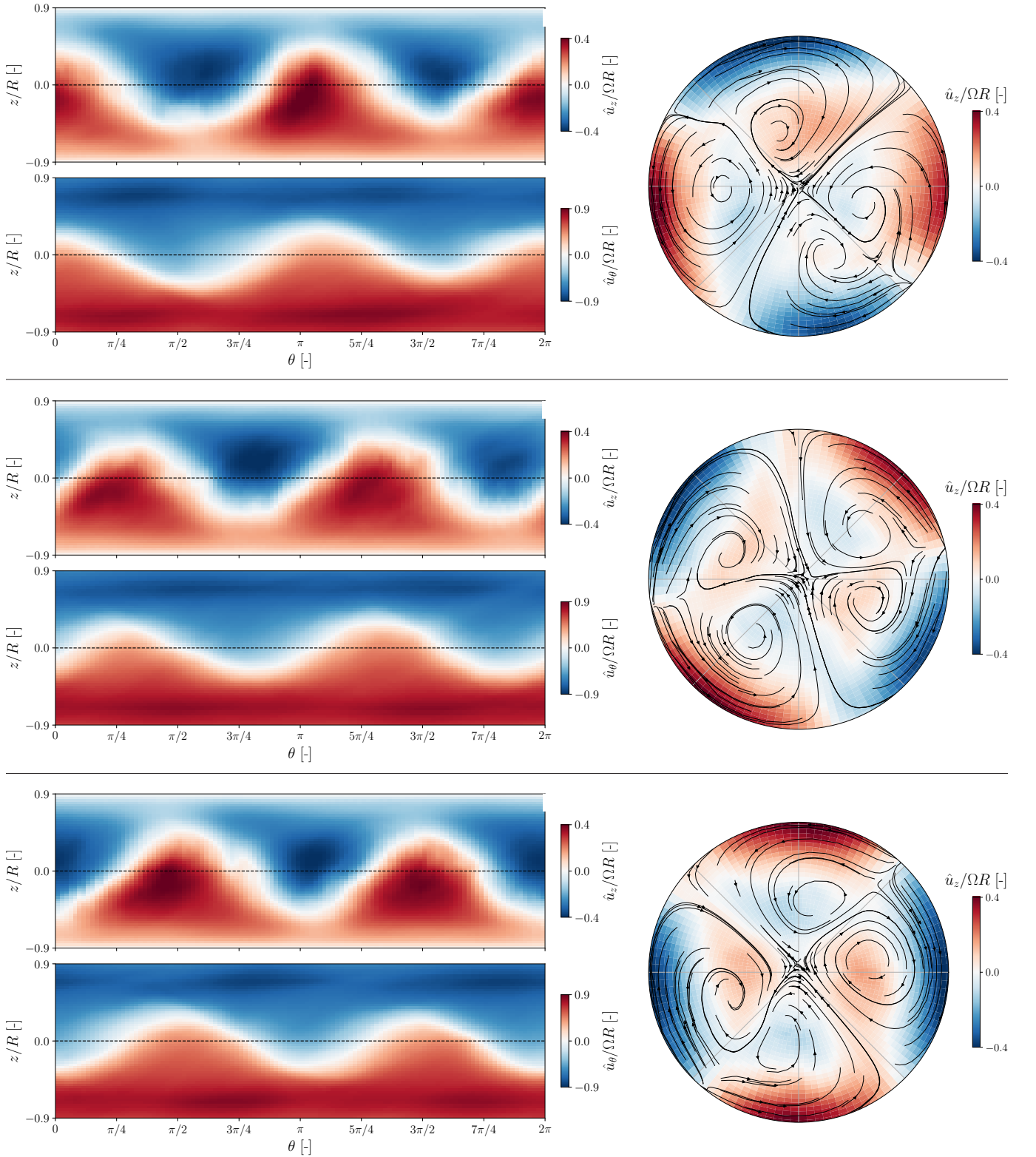


FIG. 12. Snapshots of the EMA-filtered velocity field at three distinct instants, each corresponding to a metastable state whose positive order-parameter lobe is centered at a different azimuthal position ($\theta = 0, \pi/4$ and $\pi/2$, respectively). Left: unrolled cylindrical section at the second cell from the outer wall colored by EMA-filtered dimensionless axial velocity $\hat{u}_z/\Omega R$ and azimuthal velocity $\hat{u}_\theta/\Omega R$. Right: mid-height cut colored by EMA-filtered dimensionless axial velocity $\hat{u}_z/\Omega R$, with streamlines of the in-plane $(\hat{u}_r, \hat{u}_\theta)$ velocity. A four-cell pattern is clearly visible in each mid-height cut, with axial velocity positively correlated between diametrically opposite sectors ($\Delta\theta = \pi$) and anti-correlated between sectors separated by a quarter turn ($\Delta\theta = \pi/2$).

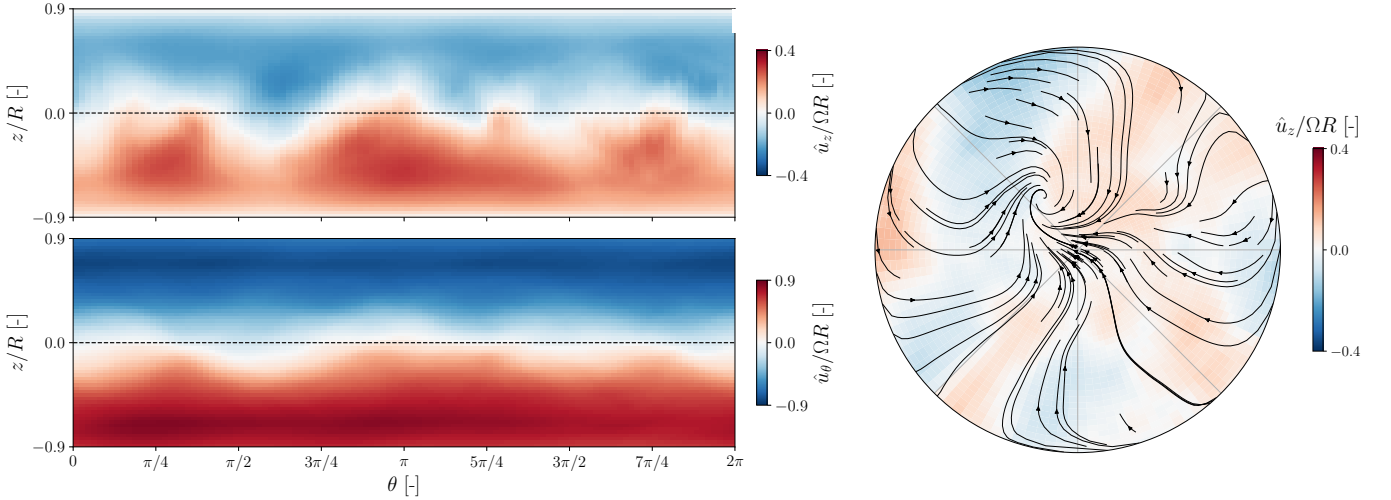


FIG. 13. Snapshots of the EMA-filtered velocity field for an instant where the state of the system is unclear. Left: unrolled cylindrical section at the second cell from the outer wall colored by EMA-filtered dimensionless axial velocity $\hat{u}_z/\Omega R$ and azimuthal velocity $\hat{u}_\theta/\Omega R$. Right: mid-height cut colored by EMA-filtered dimensionless axial velocity $\hat{u}_z/\Omega R$, with streamlines of the in-plane $(\hat{u}_r, \hat{u}_\theta)$ velocity. No four-cell structure is observed, reflecting the loss of the wavenumber-2 organization.

IV. CONCLUSION

A two-stage optimization strategy has been introduced to model impeller-driven von Kármán flow in LES without resolving blade geometry. The impeller-region velocity is expressed as a low-order B-spline parametrization whose coefficients are first determined by ODIL, which solves an inverse problem constrained by the RANS equations, experimental bulk-flow and torque measurements to recover a physically consistent representation of the impeller-induced flow without requiring direct access to the impeller-region velocities. The inferred velocities constitute a compact parametrization of the impeller forcing that is then coupled to the LES through momentum source terms. The coefficients are subsequently optimized by CMA-ES coupled directly to the LES solver, minimizing the mismatch between LES and experimental time-averaged velocity and torque. This LES-coupled optimization corrects the modeling error of the RANS-based inference and yields an improved mean-flow prediction. Crucially, the resulting LES captures metastable large-scale flow states and spontaneous transitions without imposing time-dependent forcing, enabling intrinsic instabilities characterizing the slow-regime behavior of the system to be observed. The metastable states are further shown to be spatially structured, consisting of an alternating four-cell pattern around the vessel circumference that slowly drifts in azimuth, with state transitions typically proceeding through this drift. The simulation of such flow dynamics is inaccessible to blade-resolved approaches at comparable Reynolds numbers due to their prohibitive computational cost. The proposed approach therefore provides a flexible and computationally efficient framework for studying slow, large-scale dynamics in impeller-driven turbulent flows, without explicitly resolving the blade geometry. Future work will focus on more quantitative investigation of these instabilities and their behavior for a wider range of operating conditions.

ACKNOWLEDGMENTS

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DATA AVAILABILITY

The experimental PIV and torque data that support the findings of this study are available in Ref. [15].

SUPPLEMENTARY MATERIAL

Three animations illustrate the EMA-filtered velocity field over the course of a long-duration LES, complementing the snapshots shown in Figs 8, 12 and 13.

- meridional_cut_UVWEMA_animation.mp4: animation of a meridional cut at $\theta = \pi/4$, colored by the EMA-filtered dimensionless azimuthal velocity $\hat{u}_\theta/\Omega R$, with streamlines of the EMA-filtered radial and axial velocity.
- UVWEMA_disc_cut_axial_Ucolored_animation.mp4: animation of a cross-sectional cut normal to the cylinder axis, at the mid-plane of the vessel, colored by the EMA-filtered dimensionless axial velocity $\hat{u}_z/\Omega R$, with streamlines of the EMA-filtered azimuthal and radial velocity.
- UVWEMA_unrolled_azimuth_animation.mp4: animation of the unrolled cylindrical section two cells away from the outer cylindrical wall, colored by the EMA-filtered dimensionless axial velocity $\hat{u}_z/\Omega R$ and azimuthal velocity $\hat{u}_\theta/\Omega R$.

Appendix A: Reference experimental data

1. Experimental apparatus

The reference PIV velocity fields used throughout this work were acquired by Cappanera et al. [15] on the von Kármán flow device described therein. The setup consists of a cylindrical vessel filled with water and driven by two counter-rotating impellers of TM87 geometry [31]. The dimensions of the von Kármán flow device are given in Table I. The impellers feature 8 curved blades with a blade angle of 72° . The reference data used in this work correspond to the operating point $Re = 300,000$, with the impellers counter-rotating at the same absolute frequency (i.e., $\Omega_1 = -\Omega_2$), both spinning with the concave side of the blades leading. Velocity fields in the meridional plane are measured by Stereoscopic Particle Image Velocimetry (SPIV), and the torque applied to each impeller shaft is independently measured using torque meters on the driving motors. We refer to Cappanera et al. [15] for further details on the experimental apparatus and measurement procedures.

2. Data regularization

Before being used as reference data, the mean velocity field undergoes two regularization steps: enforcement of physical symmetry and axis conditions, and a divergence-free projection. Each step is described below.

a. Midplane and axis symmetry

The von Kármán configuration with equal impeller speeds is statistically symmetric about the midplane $z = 0$, and the cylindrical mean velocity components obey definite parities:

$$\bar{u}_r(r, z) = \bar{u}_r(r, -z), \quad \bar{u}_z(r, z) = -\bar{u}_z(r, -z), \quad \bar{u}_\theta(r, z) = -\bar{u}_\theta(r, -z). \quad (\text{A1})$$

These parities are enforced by replacing each field with its symmetrized version,

$$\bar{u}_r \leftarrow \frac{1}{2}(\bar{u}_r + \bar{u}_r^-), \quad \bar{u}_z \leftarrow \frac{1}{2}(\bar{u}_z - \bar{u}_z^-), \quad \bar{u}_\theta \leftarrow \frac{1}{2}(\bar{u}_\theta - \bar{u}_\theta^-), \quad (\text{A2})$$

where the superscript $-$ denotes the field evaluated at $-z$. At the midplane row ($z = 0$) the odd components are additionally set to zero exactly: $\bar{u}_z = \bar{u}_\theta = 0$.

On the symmetry axis ($r = 0$), regularity of an axisymmetric field requires $\bar{u}_r = 0$, $\bar{u}_\theta = 0$, and zero radial gradient of $\bar{u}_z$; all three conditions are enforced explicitly.

b. Divergence-free projection

Experimental PIV data are not exactly solenoidal due to measurement noise and interpolation errors. A weighted least-squares divergence correction is applied to the pair $(\bar{u}_r, \bar{u}_z)$ to minimize the cylindrical continuity residual

$$\mathcal{D}(\bar{u}_r, \bar{u}_z) = \frac{1}{r} \frac{\partial(r \bar{u}_r)}{\partial r} + \frac{\partial \bar{u}_z}{\partial z}. \quad (\text{A3})$$

The discrete divergence is evaluated with standard second-order finite differences on the uniform (r, z) SPIV grid. A least-squares correction $\mathbf{s} = (s_r, s_z)$ is then computed by solving

$$\min_{\mathbf{s}} \|\mathbf{s}\|_2^2 \quad \text{subject to} \quad \mathbf{D}(\bar{\mathbf{u}} - \mathbf{s}) \approx \mathbf{0}, \quad (\text{A4})$$

where $\mathbf{D}$ is the discrete divergence operator and $\bar{\mathbf{u}} = (\bar{u}_r, \bar{u}_z)$ is the measured field. The system is solved with LSQR. The final corrected field is used as reference data throughout this work. A one-cell-wide boundary strip along all four edges of the domain is removed before comparison, as boundary cells carry one-sided stencil residuals that are not reduced by the interior projection.

Appendix B: Gauss–Newton algorithm for ODIL inference

The ODIL inference problem is posed as the minimization of a nonlinear least-squares objective built from a stacked residual vector $\mathbf{r}(\mathbf{v})$, which collects the discretized governing-equation residuals together with data-misfit terms (and any additional constraints). Here $\mathbf{v}$ denotes the optimization state, obtained by concatenating all unknown degrees of freedom (flow variables on the numerical grid and the spline coefficients used to parametrize the impeller-region velocities). We solve

$$\min_{\mathbf{v}} \|\mathbf{r}(\mathbf{v})\|_2^2. \quad (\text{B1})$$

At iteration k , Gauss–Newton linearizes $\mathbf{r}(\mathbf{v})$ about $\mathbf{v}^k$ and computes an update $\Delta \mathbf{v}$ by solving the system $(\mathbf{J}^\top \mathbf{J} + \lambda \mathbf{I}) \Delta \mathbf{v} = -\mathbf{J}^\top \mathbf{r}$, where $\mathbf{J}(\mathbf{v}^k) = \partial \mathbf{r} / \partial \mathbf{v}$ and $\lambda \geq 0$ is a small damping parameter ($\lambda = 10^{-8}$ in this work). The linear system is solved using GMRES by applying the operator $\mathcal{A}(\mathbf{w}) = \mathbf{J}^\top (\mathbf{J} \mathbf{w}) + \lambda \mathbf{w}$ through Jacobian–vector and transposed-Jacobian–vector products computed with automatic differentiation, without forming $\mathbf{J}$ explicitly. The step is then accepted using a backtracking line search enforcing an Armijo decrease condition on $\|\mathbf{r}\|_2$. Algorithm 1 summarizes the procedure.

Require: Initial state vector $\mathbf{v}^0$ (fields + spline coefficients), max iterations K , Armijo parameter c_1 , max line-search steps N_{ls} , damping $\lambda \geq 0$

```

1: for  $k = 0, \dots, K - 1$  do
2:    $\mathbf{r} \leftarrow \mathbf{r}(\mathbf{v}^k)$  ▷ assemble weighted residual vector (PDE + data)
3:    $\mathbf{g} \leftarrow \mathbf{J}(\mathbf{v}^k)^\top \mathbf{r}$ 
4:   Define operator  $\mathcal{A}(\mathbf{w}) = \mathbf{J}(\mathbf{v}^k)^\top (\mathbf{J}(\mathbf{v}^k) \mathbf{w}) + \lambda \mathbf{w}$ 
5:   Solve approximately  $\mathcal{A}(\Delta \mathbf{v}) = -\mathbf{g}$  using GMRES ▷ normal equations
6:    $s \leftarrow 1$ 
7:   for  $\ell = 1, \dots, N_{\text{ls}}$  do
8:      $\mathbf{v}^{\text{trial}} \leftarrow \mathbf{v}^k + s \Delta \mathbf{v}$ 
9:      $\mathbf{r}^{\text{trial}} \leftarrow \mathbf{r}(\mathbf{v}^{\text{trial}})$ 
10:    if  $\|\mathbf{r}^{\text{trial}}\|_2 \leq (1 - c_1 s) \|\mathbf{r}\|_2$  then ▷ Armijo condition on  $\|\mathbf{r}\|_2$ 
11:      break
12:    end if
13:     $s \leftarrow s/2$ 
14:  end for
15:  if  $\ell > N_{\text{ls}}$  then
16:     $\mathbf{v}^{k+1} \leftarrow \mathbf{v}^k + 10^{-2} \Delta \mathbf{v}$  ▷ fallback if no sufficient decrease
17:  else
18:     $\mathbf{v}^{k+1} \leftarrow \mathbf{v}^{\text{trial}}$ 
19:  end if
20: end for
21: return  $\mathbf{v}^K$ 

```

Algorithm 1. Gauss–Newton for ODIL residual minimization

Appendix C: Equilibrium wall model

The wall model at the top, bottom and outer cylindrical walls of the vessel is derived assuming that the near-wall boundary layer is in local equilibrium. The inner part of the turbulent boundary layer is replaced by the solution to a one-dimensional equilibrium boundary-layer problem forced by the resolved LES velocity at the first off-wall cell. Let y denote the wall-normal distance and $u(y)$ the wall-tangential velocity component in the inner layer. The mean velocity is assumed to obey the equilibrium momentum balance

$$\frac{d}{dy} \left[(\nu + \nu_T) \frac{du}{dy} \right] = 0, \quad (\text{C1})$$

where ν is the kinematic viscosity and $\nu_T(y)$ the turbulent viscosity. The wall stress reads

$$\tau_w = \rho u_\tau^2 = \rho \nu \left. \frac{du}{dy} \right|_{y=0}, \quad (\text{C2})$$

where u_τ is the friction velocity. Integrating Eq. (C1) gives

$$(\nu + \nu_T) \frac{du}{dy} = u_\tau^2. \quad (\text{C3})$$

The turbulent eddy viscosity is obtained from a mixing-length model with a van Driest damping function

$$\nu_T(y) = [D(y) \kappa y]^2 \left| \frac{du}{dy} \right| \quad \text{with} \quad D(y) = 1 - \exp(-y^+/A^+), \quad (\text{C4})$$

and with $\kappa = 0.4$, $A^+ = 25$ and $y^+ = y u_\tau / \nu$. This yields a first-order nonlinear Ordinary Differential Equation (ODE) for $u(y)$

$$\nu \frac{du}{dy} + [D(y) \kappa y]^2 \left| \frac{du}{dy} \right| \frac{du}{dy} = u_\tau^2, \quad (\text{C5})$$

subject to the boundary conditions $u(0) = 0$ and $u(\Delta_y) = U_{\text{LES},||}$, where $U_{\text{LES},||}$ is the magnitude of the wall-parallel LES velocity at the cell adjacent to the wall and Δ_y is the wall-normal distance from the wall to the center of that

cell. Equation C5 with the two boundary conditions defines a one-parameter problem whose unknown is the friction velocity u_τ .

Working in the dimensionless form $\hat{y} = y/\Delta_y$, $\hat{u} = u \Delta_y/\nu$, $\hat{\nu}_T = \nu_T/\nu$, Eq. C5 becomes

$$\frac{d\hat{u}}{d\hat{y}} + [D(\hat{y}) \kappa \hat{y}]^2 \left(\frac{d\hat{u}}{d\hat{y}} \right)^2 = \hat{u}_\tau^2 \quad \text{with} \quad \hat{u}_\tau = u_\tau \Delta_y/\nu, \quad (\text{C6})$$

and with boundary condition $\hat{u}(1) = \hat{U}_{\text{LES},||} = U_{\text{LES},||} \Delta_y/\nu = \text{Re}_\Delta$. Solving for $d\hat{u}/d\hat{y}$ yields

$$\frac{d\hat{u}}{d\hat{y}} = \frac{1}{2 [D(\hat{y}) \kappa \hat{y}]^2} \left(-1 + \sqrt{1 + 4 [D(\hat{y}) \kappa \hat{y}]^2 \hat{u}_\tau^2} \right), \quad (\text{C7})$$

so that the friction velocity is obtained as the root of

$$\int_0^1 \frac{1}{2 [D(\hat{y}) \kappa \hat{y}]^2} \left(-1 + \sqrt{1 + 4 [D(\hat{y}) \kappa \hat{y}]^2 \hat{u}_\tau^2} \right) d\hat{y} = \hat{U}_{\text{LES},||}, \quad (\text{C8})$$

for given $\hat{U}_{\text{LES},||}$. In practice, $u_\tau(\text{Re}_\Delta)$ is evaluated through the Re_τ – Re_Δ fit of Meneveau [27]. The wall stress imposed on the LES is then

$$\tau_w = \rho u_\tau^2 \hat{\mathbf{t}}, \quad (\text{C9})$$

with $\hat{\mathbf{t}}$ the unit vector aligned with the wall-parallel LES velocity at the matching cell.

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